

NOTES: CLASS-1

Periodic function:

A function $f(x)$ is said to be periodic if $f(x + T) = f(x)$ for all real x and for some positive number T . T is known as the period of the $f(x)$. Fundamental period of $f(x)$ is the smallest period of f .

Eg. $\cos x, \sin x, \sec x, \csc x$ are periodic functions with period 2π .

Result1: If T is the period of $f(x)$. Then nT is also period of f for any integer n .

Result2: The function $h(x) = af(x) + bg(x)$ has period T if $f(x)$ and $g(x)$ have period T .

Result 3: If $f(x)$ is a periodic function of period T , then $f(ax)$ with $a \neq 0$, is a periodic function with period $\frac{T}{a}$.

Result 4: The period of a sum of a number of periodic function is the least common multiple of the periods.

Result 5: A constant is periodic for any period T .

Trigonometric series is a fundamental series of the form $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$ where constants a_0, a_n, b_n are called the coefficients.

Euler's (Fourier-Euler) Formulae

Let $f(x)$, a periodic function with period 2π defined in the interval $(\alpha, \alpha + 2\pi)$, be the sum of a trigonometric series, $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$ (1)

$$\text{Then } a_0 = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) dx. \quad (2)$$

$$a_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \cos nx dx \text{ for } n = 1, 2, 3 \dots (3)$$

$$b_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \sin nx dx \text{ for } n = 1, 2, 3 \dots (4)$$

The formula (2),(3),(4) are known as Euler formulae gives the coefficient's a_0, a_n, b_n which are known as Fourier coefficients of $f(x)$.

Fourier Series

Fourier series of a periodic function $f(x)$ with period 2π is the trigonometric series (1) with the Fourier coefficients a_0, a_n, b_n given by the Euler formulae (2),(3),(4). The individual terms in Fourier series is known as harmonics.

Dirichlet conditions:

Let $f(x)$ be a periodic function with period 2π . Let $f(x)$ be a piecewise continuous, and bounded in the interval $(\alpha, \alpha + 2\pi)$ with finite number of extrema. Then

1. at the point of continuity Fourier series of $f(x)$ (RHS) converges to $f(x)$ (LHS).
2. At the point of discontinuity Fourier series of $f(x)$ converges to the arithmetic mean of left and right hand limits of $f(x)$.

Problem 1. Obtain the Fourier series to represent e^{-ax} from $x = -\pi$ to $x = \pi$. Hence derive the series for $\frac{\pi}{\sinh \pi}$

Solution : The Fourier expansion of $f(x)$ over $(-\pi, \pi)$ is given by,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \dots \dots \dots (1)$$

where,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx.$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-ax}}{-a} \right]_{-\pi}^{\pi}$$

$$= \frac{-1}{a\pi} [e^{-a\pi} - e^{a\pi}]$$

$$a_0 = \frac{2 \sinh a\pi}{a\pi}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} \cos nx dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-ax}}{(-a)^2 + n^2} (-a \cos nx + n \sin nx) \right]_{-\pi}^{\pi}$$

$$\left\{ \text{Since, } \int e^{ax} \cos bx dx = \left[\frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) \right] \right\}$$

$$= \frac{-a}{\pi(a^2 + n^2)} [e^{-a\pi} \cos n\pi]_{-\pi}^{\pi}$$

$$= \frac{-a \cos n\pi}{\pi(a^2 + n^2)} (e^{-a\pi} - e^{a\pi})$$

$$a_n = \frac{2a(-1)^n \sin h a \pi}{\pi(a^2 + n^2)}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} \sin nx \, dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-ax}}{(-a)^2 + n^2} (-a \sin nx - n \cos nx) \right]_{-\pi}^{\pi}$$

$$\text{Since } \left\{ \int e^{ax} \sin bx \, dx = \left[\frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) \right] \right\}$$

$$= \frac{-n}{\pi(a^2 + n^2)} [e^{-ax} \cos nx]_{-\pi}^{\pi}$$

$$= \frac{-n \cos n\pi}{\pi(a^2 + n^2)} (e^{a\pi} - e^{-a\pi})$$

$$b_n = \frac{2n(-1)^n \sin h a \pi}{\pi(a^2 + n^2)}$$

Therefore the required Fourier series is given by,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$e^{-ax} = \frac{1}{2} \frac{2 \sin h a \pi}{a \pi} + \sum_{n=1}^{\infty} \left(\frac{2a(-1)^n \sin h a \pi}{\pi(a^2 + n^2)} \cos nx + \frac{2n(-1)^n \sin h a \pi}{\pi(a^2 + n^2)} \sin nx \right)$$

$$\Rightarrow e^{-ax} = \frac{\sin h a \pi}{a \pi} \left\{ 1 + \sum_{n=1}^{\infty} \left(\frac{2a^2(-1)^n}{(a^2 + n^2)} \cos nx + \frac{2an(-1)^n}{\pi(a^2 + n^2)} \sin nx \right) \right\}$$

Deduction :

To deduce the series we shall put $a = 1, x = 0$ in the above equation :

$$e^0 = \frac{\sin h \pi}{\pi} \left\{ 1 + \sum_{n=1}^{\infty} \left(\frac{2(-1)^n}{(1 + n^2)} \cos n(0) \right) \right\}$$

$$\Rightarrow 1 = \frac{\sin h \pi}{\pi} \left\{ 1 + \sum_{n=1}^{\infty} \left(\frac{2(-1)^n}{(1 + n^2)} \right) \right\}$$

$$\Rightarrow \frac{\pi}{\sinh \pi} = 1 + 2 \left(\frac{-1}{1+1^2} + \frac{1}{1+2^2} + \frac{-1}{1+3^2} + \frac{1}{1+4^2} + \dots \right)$$

$$\Rightarrow \frac{\pi}{\sinh \pi} = 1 - 1 + 2 \left(\frac{1}{1+2^2} - \frac{1}{1+3^2} + \frac{1}{1+4^2} - \dots \right)$$

$$\Rightarrow \frac{\pi}{\sinh \pi} = 2 \left(\frac{1}{1+2^2} - \frac{1}{1+3^2} + \frac{1}{1+4^2} - \dots \right)$$

REFERENCES:

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